SOURCES OF FINANCE

13

A LION RETREATS

On 25 September 2023, AngloGold Ashanti plc moved the group's primary listing from the Johannesburg Stock Exchange (JSE) to the New York Stock Exchange (NYSE). Its corporate domicile had previously moved to the UK, yet the group would retain a secondary listing on the JSE and the A2X. AngloGold is one of the world's largest gold mining companies and has been a leading presence in the South African mining sector. Its logo is a lion which reflects its deep roots in Africa, particularly South Africa. Why move their primary listing to New York? In this chapter we will evaluate reasons for moving a company's primary listing. However, AngloGold Ashanti also offers lessons in equity financing by using rights issues to repay borrowings and finance operations. On 10 September 2014, AngloGold announced that it was making a $2.1 billion rights issue (an offer to current shareholders to subscribe for shares) in order to obtain equity capital to reduce its net debt of about $3 billion. The rights issue amounted to one third of its market capitalisation. The group also planned to split its South African assets from its international assets. The market reaction to the announcement was swift and dramatic, with the share price falling from $15.49 (R169) to $13.07 (R144) on the day. Most fund managers and shareholders voiced concern and indicated that they would not support the rights issue, stating that this would be dilutive to shareholder value (which is just a polite way to say that it was expected to destroy value). Furthermore, the first repayment date for the debt (bonds) was only 2020 and there was no immediate need to raise equity capital. AngloGold quickly retreated and cancelled the rights issue and indicated that it would engage in asset sales, reduce costs and enter into partnerships on some projects to generate savings to repay debt. By 2018, the group continued to flounder with a share price at $9.59 (R138). In 2023, the group is trying to improve operations and extend the life of its mines. Operating profit of AngloGold Ashanti fell from $1497m in 2020 to $519m in 2022 and its USD share price in 2023 was a third of its value in 2006. The situation was so different in 2008 when AngloGold Ashanti made a R13.48 billion (US$1.77

billion) rights issue, which was almost 7 times oversubscribed. How times change! The rights issue price was then set at R194 per share when the share price at the time was R264. Why would AngloGold issue shares at a discount to its share price? This is one of the many aspects we will explore in this chapter.

LEARNING OBJECTIVES

After working through this chapter, you should be able to:

- Define the different types of financial markets.
- Explain the role of the JSE (Johannesburg Stock Exchange) in raising finance.
- Detail the listing criteria for the JSE.
- Understand the valuation of rights.
- Detail the types of financial institutions.
- Explain the types of financial instruments.
- Distinguish between equity and debt-related instruments.
- Understand how credit ratings work.
- Explain criteria by which to evaluate sources of finance.
- Understand black economic empowerment (BEE) financing structures.

INTRODUCTION

Every company needs finance to conduct its operations. The financial manager should consider the broad categories of finance to be raised, the mix of debt and equity, and the cost of the finance. In addition, the financial instruments that will be issued to raise the finance must appeal to investors otherwise funds will not be forthcoming. Finance can be supplied from various sources, with terms and conditions to satisfy the needs of particular investors, and to produce the risk/return relationship required. In this chapter we will identify the markets and the institutions through which finance can be raised as well as the types of finance available, and we will compare debt and equity finance.

1 FINANCIAL MARKETS

In the market place where everyday commodities are traded, sellers have goods to sell and buyers are looking for goods to purchase. If the terms are agreed upon, a transaction is concluded and goods change hands at a price. In much the same way, finance is traded in the market place. There are well-developed markets for the trading of financial assets.

Classification of financial markets

Some useful and common classifications of the financial markets are described below.

Money and capital markets

The money market is used mainly for raising short-term finance. The money market is not situated in any one physical location and consists mainly of groups of interested parties who borrow and lend amongst each other. Institutions such as merchant banks, commercial banks, non-bank financial institutions and large corporations operate in the money market.

The capital market is used mainly for the raising of long-term finance through, for example, the JSE. Like so many distinctions in finance, the separation of markets into money and capital markets is really a guideline and not a rigidly applied distinction. As a rule of thumb, money-market transactions are those dealing in financial instruments with a life of less than three years, while the capital market is for instruments with a life greater than three years.

Some instruments clearly fall into one or the other of the categories. For example, the shares of a company listed on the stock exchange fall into the capital market, while 90-day bankers' acceptances are clearly money-market instruments.

Primary and secondary markets

The primary market is the market for new issues of finance, whereas the trading of securities already issued takes place in the secondary market. Shares bought and sold as daily transactions on the stock exchange are an example of dealings in the secondary market. However, an offer of new equity on the JSE is an example of a primary market transaction.

The secondary market is not a source of finance, but it is nevertheless relevant as it indicates the marketability and value of the securities. In this way it provides information for the organisation wanting to raise finance in the primary market.

Formal and over-the-counter (OTC) markets

Formal markets, such as exchanges, are highly regulated; they usually specialise in a narrow range of instruments; and they are accessible to small investors. Deals on the OTC market are carried out across the desks of interested parties using computer terminals and telephones.

Most exchanges started out as OTC markets and were formalised as the need arose. However, OTC markets do not necessarily become formal markets, as there is a need for some markets to remain informal. Standardised instruments are traded on formal markets, whereas the OTC markets permit the parties to innovate in accordance with their needs.

Moreover, certain standardised instruments do not need formal exchanges to trade efficiently. For example, currencies are efficiently traded on the OTC market, usually using terminals and telephones. Examples of formal exchanges in South Africa are the JSE Equity Market, the A2X, the JSE Derivatives Market (previously known as SAFEX), which is a division of the JSE, and the JSE Debt Market [(previously known as the Bond Exchange of South Africa (BESA)] which is also a division of the JSE. Page 13-3

Spot and derivative markets

In the spot market, instruments or commodities are traded at an agreed price immediately, whereas in the case of derivative instruments, value is derived from the price of an underlying instrument or commodity.

For example, an option to buy BHP shares at R580 represents a derivative contract as the value of the option will be linked to or derived from the value of the underlying asset, which is a BHP ordinary share. A futures contract to buy US dollars at R19.20 in 90 days' time is an instrument that is derived from the value of the underlying asset, in this case being the value of the US dollar today.

The primary purpose of the derivatives market is to manage risk. However, the flexibility, leverage and low cost of derivative instruments also encourage their use by investors. Derivative contracts can be exchange traded or non-exchange traded.

Exchange traded derivative instruments are standardised contracts traded on physical exchanges and include options and warrants on the JSE and futures and options on futures on the JSE Derivatives Market (SAFEX).

In contrast, non-exchange traded derivative instruments are non-standardised (negotiated independently between the parties), and are not traded on a physical exchange (therefore less liquid than exchange traded derivatives). Examples include forward contracts, forward rate agreements (FRAs) and swaps.

Interaction between market classifications

The classifications of markets described above are not mutually exclusive. For example, if you sold 1 000 MTN shares, this transaction could be classified into the following markets:

- capital – the shares are long-term instruments;
- secondary – you get the money and the buyer gets the shares (MTN is not involved);
- spot – you get the money and deliver the scrip immediately (this is done electronically via STRATE); and
- formal – you sold your shares through your broker who traded them on the JSE.

The Johannesburg Stock Exchange

The Johannesburg Stock Exchange (JSE) was founded by Benjamin Woollan in November 1887. Many mining and financial companies were formed after the discovery of gold on the Witwatersrand (Gauteng). A market place was needed for the transaction of the shares in those companies, and the JSE was formed to provide such a market place. The JSE, up to recently, was the only stock exchange in South Africa. As from 2017, the A2X went live and is South Africa's second stock exchange. There are also the CTSE and EESE exchanges. Other countries also have more than one stock exchange. For example, in the United States of America there is the New York Stock Exchange, NASDAQ and the American Stock Exchange. The JSE is governed by an Act of Parliament, the Financial Markets Act, 2012, and by its own rules and directives. The rules and directives include requirements concerning listings, trading, disputes, brokerage, settlements and defaults, among other items. Trading on the JSE was in the form of an auction involving brokers. The brokers called out the names of the shares for which they had orders to buy or sell. The prices of transactions were recorded on a price board which was situated in the market place. In 1999, the JSE introduced the STRATE project (Share Transactions Totally Electronic) which allows transactions to be settled and traded electronically. This means that users are assured

that trades will be settled in full and on due date. By February 2002 all JSE counters had moved to STRATE. Equity scrip is no longer in the form of paper share certificates. STRATE results in the dematerialisation of share certificates and there is a Central Securities Depositary which holds the electronic equity 'certificates'. Page 13-4

The main types of securities listed on the JSE and its divisions are:

- Ordinary shares;
- Preference shares;
- Derivatives (warrants, futures & options);
- Bonds & Notes.

Futures and options are traded on the JSE Derivatives Market (SAFEX), which is a division of the JSE. These are also called derivatives as their value is derived from an underlying security or commodity. The JSE Derivatives Market has financial derivatives, currency futures and options and agricultural futures and option contracts over such commodities as wheat and maize.

Ordinary shares represent the ownership of companies and are by far the most important securities listed on the JSE. **Preference shares** pay a fixed dividend rate and often are redeemable at the issue price. We will come back to warrants, futures and options later in Chapter 18, and a brief description of each security will suffice for now. An option is the right but not the obligation to buy the underlying share at a specified price within a specified period. This is a *call option*. The right to sell at a specified price is called a *put option*. Warrants represent options on the ordinary shares of listed companies and are often issued by banks such as Standard Bank who have hedged their positions by buying the underlying shares. There is often a conversion factor so that 5 warrants may enable you to acquire 1 ordinary share. A futures contract is an agreement to buy or sell a specified quantity of an asset or commodity at a fixed price at a specified time in the future. There are also other listed instruments such as instalments created by the banks which mean that you invest only part of the amount required to acquire a specified number of shares.

What about debt securities? Corporate, parastatal, municipal and government bonds are listed on the JSE Debt Market, a division of the JSE. We will describe the role of the JSE Debt Market (BESA) later in the chapter.

What is the role of the JSE? A JSE listing enables companies and management to:

- issue shares and raise primary capital to finance business expansion;
- ensure there is a secondary market for the trading of shares so that investors can buy and sell securities;
- monitor the performance of the company as measured by movements in its share price;
- offer employees share incentives and share options;
- enhance the public profile of the company;
- borrow loan finance from the banks at improved rates;
- improve the disclosure to its shareholders and ensure that the market is aware of developments, operations and financial results;
- finance the acquisition of other unlisted or listed companies by issuing shares in the company to the selling shareholders; and
- enter into share buy-backs if the company does not require further capital.

It is important for the JSE to ensure that there is a liquid market, that it offers protection to investors in the form of fair and equal treatment, and that there is proper and timely disclosure.

How effective has the JSE been in enabling companies to raise primary capital? Figure 13.1 shows the sums of new equity capital raised annually on the JSE since 1998. Page 13-5

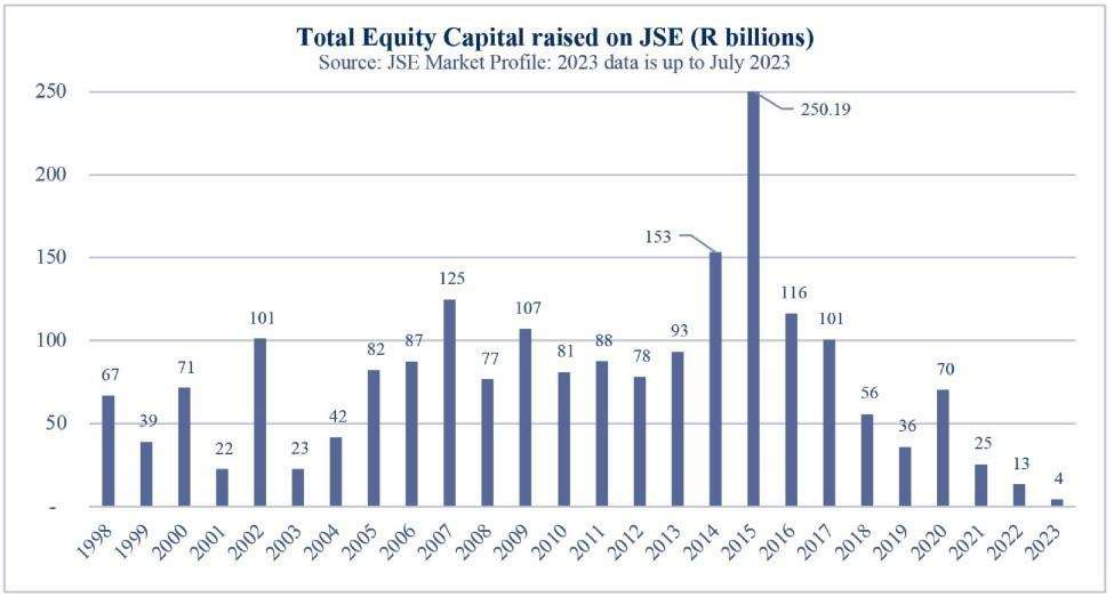


Figure 13.1 New equity capital raised on the JSE (R billions)

Source: JSE Market Profile, SA Reserve Bank Quarterly Bulletin

We can see that the performance of the JSE has been variable in terms of raising primary capital and in real terms, it has not been impressive to date. In recent years, the JSE's role in raising equity financing has significantly weakened and this is in nominal terms, not in real terms. The R67bn equity funding raised on the JSE in 1998 would be the equivalent of R226bn in 2023 if we apply a reasonable inflation rate of 5% per year. Yet, total equity raised in 2022 was only R13bn and it will be lower in 2023. In fact, the number of listed companies has fallen dramatically in recent years as companies have gone private and minority shareholders have been bought out, mainly via schemes of arrangement.

Table 13.1 Number of listed companies on the JSE (domestic)

Mainboard*	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	201
No. of New Listings	74	14	11	9	8	17	19	37	62	23	10	14	16	14	13	18
No. of Delistings	75	66	85	79	54	39	35	24	40	20	25	17	17	18	26	20
Mainboard* &AltX	**1999**	**2000**	**2001**	**2002**	**2003**	**2004**	**2005**	**2006**	**2007**	**2008**	**2009**	**2010**	**2011**	**2012**	**2013**	**201**
No. of listed companies#	668	616	542	472	426	404	388	371	385	379	363	360	355	350	333	32

**Mainboard, Development Capital Market, Venture Capital Market & BEE # Domestic companies. There were 60 Market Profile: 2023 data is up to July 2023]*

Companies that have delisted include such companies as Distell, Massmart, Pioneer Foods, Mediclinic, I&J, McCarthy, Cadbury, Schweppes, Nando's, M-Net, Safren, Dimension Data, Afgri, SABMiller, Pikwik, Group Five and LTA. New listings over the last 20 years include companies such as Spar, Kumba Resources, Vodacom, Adcock Ingram, Sibanye, Glencore, Dis-chem, Sea Harvest, Kaap Agri, AYO, South32, AB InBev, Pepkor and Alexander Forbes. Listed companies have undertaken major share buy-backs in effect giving back capital to shareholders.

Although listing on the JSE does have major benefits, the fact that companies have been delisting implies that there are also disadvantages to listing. These include the following:

- Disclosure requirements and reporting standards can be onerous requiring a greater investment in information systems, and resulting in an increase in compliance

costs.

- Increased accountability to shareholders and loss of privacy and flexibility in managing the company. Page 13-6
- Financial information about margins and profitability levels become available to competitors and potential competitors. Any strategic information shared with investors will be accessed by competitors. Suppliers and customers become aware of the financial performance of the company.
- Management time is an issue, particularly for smaller companies, as the management time involved in the listing process, reporting to the market, financial reports, interaction with analysts and financial institutions may take time away from managing the business.
- Increased attention on the company may also be negative if the company reports difficult trading conditions and the share price falls.
- Costs of listing, annual fees and continuing costs relating to public relations and investor relations.
- Increased responsibilities on directors and increased focus on corporate governance. Directors are required to disclose remuneration levels and share dealings and are required to communicate to the market if price-sensitive information is presented to them.
- There is a lack of market liquidity particularly for the smaller listed companies. This results in a lack of institutional interest in these companies. Furthermore, there is limited research coverage of the small and mid-cap listed companies.

The directors are required to apply the King IV Code on Corporate Governance, the listing requirements of the JSE, the Companies Act 71 of 2008 (which includes the Takeover Regulations), the Financial Markets Act 19 of 2012 and other legislation that applies to the company's industry sector. However, the fact is that many companies would not be able to expand business operations without raising equity capital on the JSE or having a secondary market for investors to trade in shares.

The JSE operates two markets, the Main Board and AltX. Although, currently, there are also the Development Capital Market (DCM) and the Venture Capital Market (VCM), it is expected that these two markets will close at some future time and companies will transfer to the AltX or the Main Board.

Listing requirements for the Main Board of the JSE include the following:

- A minimum existing subscribed capital of R50m which includes reserves and retained earnings but excludes minority interests and asset revaluations. An independent valuation by a JSE approved expert is required in order to include intangible assets.
- At least 25 million shares in issue.
- A satisfactory audited profit history for the preceding three years, the last of which reported an audited profit of at least R15 million before taxation or an existing subscribed capital of at least R500m which includes reserves and retained earnings but excludes minority interests and asset revaluations. An independent valuation by a JSE approved expert is required in order to include intangible assets.
- At least 10% of each class of equity securities are held by the public which represents at least 100 shareholders.

Companies are required to comply with other requirements such as the adoption of international financial reporting standards, reporting on price-sensitive information, reporting of related party transactions, working capital reports, corporate governance, timing of providing financial statements and the appointment of a sponsor.

For companies delisting from the Main Board, it is now a requirement that such delisting occurs at fair value and there is approval by the non-controlling shareholders. Page 13-7

Alternative Exchange – AltX

The JSE established in 2003 a new market for small to medium-sized companies, called the Alternative Exchange (AltX). The objective is to make listing less onerous for small to medium-sized companies that are currently experiencing a high growth rate and may transfer their listing to the Main Board at a later stage. The listing requirements for the AltX are less onerous than for the Main Board but the AltX is determined to ensure that all listings onto the AltX represent high quality listings. In July 2023, there were 30 companies listed on the AltX. Up to 2018, 141 companies had listed on the AltX and more than R48.5bn was raised. Over 30 companies have migrated from the AltX to the Main Board and the balance of companies have delisted. Well-known delistings included 1Time Airline and Country Foods.

Companies that moved from the AltX to the Main Board include, Curro, Ellies, Finbond and Litha. The AltX is a separate division of the JSE.

Listing requirements for the AltX include the following:

- Share capital of at least R2 million which includes reserves but excludes minority interests and asset revaluations. Intangible assets will be excluded unless supported by a JSE approved independent valuation.
- Public ownership of at least 10% of each class of equity securities to ensure liquidity.
- The number of public shareholders has to be at least 100.
- The company is required to appoint a Designated Advisor (DA). The DA undertakes a due diligence to determine whether listing on the Altx would be appropriate and provides guidance on the methods of listing. Furthermore, the DA will manage the application and listing process and will ensure all documentation is correctly completed and in line with the listing requirements. The DA ensures that the prelisting statement is in line with the listing requirements.
- The company is obliged to appoint an executive financial director and the audit committee is required to confirm to the JSE that such a financial director has the relevant qualifications and experience to undertake the role of financial director.
- Profit forecasts are required for the year of listing and the subsequent year. Historical financial information is required for the three prior years.
- Directors are required to have completed the AltX Directors' Induction Programme.

The DA reviews financial information announcements with the directors to ensure accuracy and full disclosure and undertakes regular reviews of the financial performance and financial position of the company. The DA is required to attend board meetings.

There is no requirement of a minimum level of profitability or the existence of a profit history. There are also further requirements such as holding 50% of directors' shares in trust for a minimum period.

The performance of the AltX has been disappointing and a lack of liquidity remains a significant challenge, which means it is not easy to trade shares in AltX companies. In addition, there are many companies, which are closely held by the directors,

thereby restricting the number of shares that are available to trade. The financial institutions generally will not invest in shares that are not liquid. Very few companies have delisted due to bankruptcy. Delistings are mainly due to acquisitions and other transactions. For example, Grindrod acquired majority control of Racec, which it delisted from the AltX.

The Development Capital Market (DCM) and the Venture Capital Market (VCM)

The DCM and the VCM were developed to enable the listing of small to medium sized companies. The functions of the DCM and the VCM have been taken over by the AltX. Over time, fewer and fewer companies will remain listed on either the DCM or the VCM. Page 13-8

A2X Markets and Other Exchanges

In April 2017, South Africa's second exchange, A2X, went live with three listings: Coronation, Peregrine and African Rainbow Capital. By November 2018, companies such as Standard Bank, Growthpoint, Sanlam, Anchor, AVI and Ascendis Health had listed on the A2X. At the time of writing in 2023, the A2X is only available for secondary listings and most companies will have a primary listing on the JSE. The A2X is an alternative place to transact in securities at a lower cost. Currently, companies will not be able to raise capital on the A2X and the A2X is not able to accommodate initial public offerings. However, this is expected to change in the future. The use of advanced exchange technology has resulted in lower transaction costs in the order of a 50% reduction in fees and has resulted in a narrowing of bid/offer spreads. It has led to lower friction costs and improved liquidity. In August 2023 there were 169 instruments listed on the A2X with a market capitalisation of over R10 trillion. This represents significant growth and A2X is providing the JSE with real competition. It is forecast that the A2X could provide investors with up to R1 billion per year in savings. Other exchanges or trading platforms that were formed since 2017 include ZAR X (closed in 2023 due to a lack of capital), Cape Town Stock Exchange (CTSE) [previously known as 4AX] and Equities Express Securities Exchange (EESE). Whilst these exchanges provide alternative listing platforms to the JSE, A2X acts only as a secondary market for now but we think that this is about to change. The CTSE is focused on listing small and medium-sized businesses, as well as providing services to existing BEE and other schemes such as restricted trading

schemes. It also provides a secondary market. EESE is focused on broad based BEE schemes, enabling liquidity and managing restrictions on trading. It provides direct market access.

Alternative methods of obtaining a listing

How does a company that is unlisted go about getting a listing? Essentially there are four ways: an introduction, a private placing, a public offer and using a cash shell that is already listed on the JSE.

Introduction

If the company already has the requisite number of public shareholders and does not need to raise capital, then the company can acquire a listing right away without having to make an offer to the public. This is a low cost option for the company.

Private placing

In this case, ordinary shares are sold directly to institutional investors. Usually this is done by means of the investment bank and the company's management undertaking a 'road show' and making presentations to financial institutions such as Sanlam and Old Mutual, who may then subscribe for shares. No underwriting is required. It is low cost and no prospectus is necessary. It also results in a strategic allocation of shares and the institutions may subscribe later for more shares if required.

Public offer

A public offer will be either an *offer for subscription* or an *offer for sale*. In an *offer for sale*, existing shareholders sell shares to the public and the proceeds go to the current shareholders. In an *offer for subscription* the company invites the public to subscribe for shares that have not yet been issued and the proceeds go directly to the company to finance its operations and investment opportunities. This is also widely known as an Initial Public Offering (IPO). In a public offer, the company will be required to produce a pre-listing statement or prospectus.

'Back door' listing

What is a 'back door' listing? In this case a company obtains a listing indirectly by using an existing listed company to achieve a listing. This is often undertaken by using a cash company *(cash shell)*, which is a listed company whose only asset is

cash as the company has sold off its other assets. There are two alternative ways of achieving this – either by selling the company's business to the cash shell in exchange for shares, or by arranging that the shell company takes over the operating company and is required to issue so many shares that its current shareholders lose control and the new shareholders acquire control of the listed company. This is called a reverse takeover and we will come back to this when we analyse mergers and acquisitions in Chapter 17.

Special Purpose Acquisition Company (SPAC)

The JSE permits an experienced management team with mergers and acquisitions experience or operating experience to list a SPAC which the JSE defines as "a publicly traded shell that has a specific mandate to acquire or merge with other companies or assets". At the time of listing the company (SPAC) will have no operations and consists of a "clean public shell". If acquisitions are not implemented within 24 months, then the capital is returned to the shareholders.

Raising capital by listed companies

When an existing listed company needs to raise further equity capital, then the company may make a *rights issue* to its current shareholders who may or may not subscribe for the shares. If current shareholders do not take up the right to subscribe for further shares in the company, then their holdings will be diluted. Each right will have a value. We will value such rights later in the chapter.

Setting an issue price

How does a company go about setting a listing price for its shares? The setting of an issue price is a critical part of the listing process and requires that the company's management and the corporate advisor undertake a valuation of the company's shares. If the shares are underpriced, then the current shareholders lose. If you are the sole shareholder and own 10m shares in a company and you wish to sell 3m of your shares to the public at R3.00 per share, when the real value is R5.00 per share, then you will lose R6m (R2 × 3m). Yet it seems that companies often underprice initial public offerings. Studies for South Africa up to 1998 indicated that the average first day increase in the post-listing price was 27–32%. The average increase in price on listing in the USA was 15%. For the AltX, it was found that the average increase in price was 29% up to 2007 while for the Main Board of the JSE, the average increase in the price

on listing had fallen to 10.1% for the years 2000 to 2011. Underpricing occurs mainly in IPOs of smaller companies. Why does this occur? Firstly, if there is a bargain price for an issue, then such an issue will be oversubscribed and you may receive only a small part of what you asked for. For example, assume that A Ltd is offering 10m shares at R2.00 each when their real worth is R3.00. You subscribe for 1m shares and send in your cheque for R2m. Unfortunately there are many investors like you, the share issue is oversubscribed, and so you receive only, say, 100 000 shares. You receive a cheque back after a while for R1.8m. Well, you have still made a profit of R100 000 on this issue, although you lost some interest. You then subscribe for B Ltd which is also offering 10m shares at R2.00 each but the shares are worth only R1.00 each. Unfortunately, because the expert investors and insiders know this and you don't, you get all the shares you subscribed for, which are 1m shares, and you quickly lose R1m. This represents what is sometimes called the 'winner's curse'. Therefore IPOs have to offer returns to induce investors to buy shares in the listing company. The JSE's own guidelines indicate that companies should underprice their issues by about 20% to ensure that a share issue is successful. Page 13-10

Rights issue

A company may invite existing shareholders in a company to buy more shares in that company through a rights issue. This allows the company to raise more capital, and if all the shareholders exercise their rights, then their proportionate ownership of the company remains unchanged. In most cases the subscription price to the rights issue will be set lower than the current share price to encourage existing shareholders to take up the offer to buy more shares.

Example:

A company wishes to raise R50m through a rights issue. The shares are currently trading at R6.00 and the subscription price has been set at R5.00. There are currently 90m shares in issue.

To raise R50m, the company will have to issue 10m new shares.

10 000 000 × the subscription price of R5.00 = R50 000 000

Each shareholder will then be entitled to a proportion of these new shares. A shareholder with an existing shareholding of 900 shares would be entitled to 100 new shares under the rights issue:

$$10\,000\,000 \times \frac{900}{90\,000\,000} = 100 \text{ shares}$$

The rights ratio is the number of new shares issued for each existing share held:

$$\text{Rights ratio} = \frac{10\text{ m}}{90\text{ m}} = 1{:}9$$

Shareholders will require 9 existing shares to acquire 1 new share, therefore the issue will be a 1:9 rights issue. To determine the theoretical value of each share after the rights issue we need to divide the market value of the company after the issue by the total number of shares that will then be in issue:

	R
90 000 000 × R6.00	= 540 000 000
10 000 000 × R5.00	= 50 000 000
	= 590 000 000

Theoretical value of shares after the rights issue:

$$= \frac{590\text{ m}}{100\text{ m}}$$

$$= \text{R}5.90$$

The theoretical post issue price of R5.90 is lower than the current share price but substantially higher than the subscription price of R5.00. If the rights issue is non-renounceable the shareholders may exercise the rights themselves or decline the offer. If the issue is renounceable the shareholders can exercise the rights themselves or transfer their rights to a third party. The value of the right will depend on the theoretical value of the shares after the issue, the subscription price and the number of shares required to buy one new share.

Theoretical share value post issue	= R5.90
Subscription price	= R5.00
Value of right	= R0.90

 Shares sold prior to the date at which the right to take up further shares through the rights issue are no longer attached to them are sold cum-rights. Shares sold after this date are sold ex-rights.

$$\text{Value of right: } = \left[\frac{Mc - S}{1 + R}\right]$$

where: Mc = share price cum-rights

S = subscription price

R = rights ratio

$$\left[\frac{\text{R}6.00 - \text{R}5.00}{1 + 1/9}\right]$$

Value of right = R0.90

Alternatively, the value of a right may be restated as follows:

$$\text{Value of right} = \left[\frac{N(Mc - S)}{N + 1}\right]$$

where: N = number of shares required to acquire 1 right

$$= \left[\frac{9(6.00 - 5.00)}{10}\right]$$

$$\text{Value of right} = R0.90$$

The obvious advantage to shareholders is that they can acquire further shares at a discount to the theoretical value of the shares ex-rights if the subscription price is set below the current share price. Shareholders should also consider the possibility the ex-rights share price could trade much lower than this figure as the decision by management to issue rights as a means of raising further capital can signal that the company's shares are currently overvalued and lead to a downgrade of the listed price of the security by the market.

Some facts about market liquidity

The capacity of the stock exchange to provide primary equity capital is dependent on the ability of the market to offer investors a high level of liquidity as a secondary market. This means that investors are able to buy and sell a large number of shares very quickly at the current price.

Figure 13.2 Market Liquidity of the JSE (Annual turnover as % of Market Capitalisation)

 The introduction of an automated trading system, low trading costs and other developments resulted in a significant increase in the liquidity of the JSE as indicated by the percentage of turnover to market capitalisation. The liquidity ratio increased from 5% to 40% over 23 years to 2018 but has been falling in recent years.

Liquidity remains a problem for the JSE, particularly for smaller listed companies. The top 40 listed companies make up 90% of the market capitalisation of the JSE and these are the companies that are most liquid. In Chapter 4, we indicated the effects of low levels of liquidity ('thin trading') on the estimation of betas and we gave examples of thinly traded shares. The JSE in July 2023 was ranked 19th in the world in terms of market capitalisation but only 31st in terms of liquidity. According to *Trading Economics*, the turnover ratio for the USA was 109% and 71% for Australia whilst Germany and Brazil had turnover ratios of 79% and 87% respectively. Low levels of liquidity result in investors not being able to trade shares and may mean that companies may be less willing to list on the JSE. In fact, the low level of liquidity may be one of the reasons that the number of domestic companies listed on the JSE (Main Board) has fallen from 668 in 1999 to 234 in July 2023.

The JSE Derivatives Market

The JSE Derivatives Market (previously known as SAFEX) is a market for the trading of derivatives such as futures and options. The JSE Derivatives Market trades financial and agricultural derivatives and there are futures and options contracts on wheat and maize. Financial futures and options contracts include futures contracts on indices such the ALSI40 index, other indices, interest securities and on individual highly traded shares. The JSE Derivatives Market enables the hedging of underlying positions and the volumes on the JSE Derivatives Market significantly exceed the volumes on the Main Board. We will come back to the JSE Derivatives Market in Chapter 18.

The JSE Debt Market

In 2009, the JSE acquired the Bond Exchange of South Africa (BESA), which became a division of the JSE and is now known as the JSE Debt Market or the Interest Rate Market. The JSE Debt Market had over 1500 listed debt instruments, with a total nominal outstanding amount of over R5 trillion in 2023. Most of the debt instruments are either South African government treasury bonds or debt instruments issued by state entities such as Eskom and municipalities. The R186 bond had a total turnover of

R6.8 trillion in 2016 which is a turnover ratio of 39× which is 3900% as compared to 40% at the time for the JSE. Listed companies are issuing corporate bonds or preference shares that are listed on the JSE. However, liquidity is concentrated in a few government bond issues and there is a lack of liquidity in relation to corporate bonds. Companies such as Barloworld, Telkom, MTN and Sasol have issued corporate bonds or notes. In 2018, the value of instruments traded daily was R74 billion and yet only about R8m of corporate bonds were traded daily. There is a highly active and liquid Repo market, which is geared to provide short-term financing to firms and banks. We explained what a "repo" is in Chapter 12 - it represents a short-term source of finance for companies and banks. In Chapter 6, we undertook the valuation of a listed corporate bond issued by Eskom. Companies are now able to reduce the cost of debt financing by issuing bonds and listing the bonds rather than borrowing from the banks. The use of securitisation and asset-backed securities by companies such as Edcon, SA Home Loans and Woolworths in prior years has been facilitated by having a market to list these securities. However, the financial crisis led to a decline in the number of securitisations. We will come back to explain securitisation in Chapter 18. What kind of debt securities are listed on the JSE's debt market?

- Fixed interest rate bonds
- Floating rate notes (variable interest rate bonds)
- Zero coupon bonds
- Inflation linked debt securities Page 13-13
- Commercial Paper
- Credit linked notes
- Asset backed securities
- Mortgage backed securities

Debt securities listed on the JSE in 2017 related mainly to government (65.7%), state-owned companies (12.4%) and financial institutions (16.6%). Fixed rate securities represent 69.8% of listed debt securities on the JSE while floating rate notes amount to 14.2% and inflation- linked debt securities amount to 14.5%.

Interest on government bonds is normally payable semi-annually. The bond market is very liquid with trading volumes of about R27 trillion per year which equates to

over R74 billion per day. In comparison, the volume on the JSE's Main Board is about R17 billion per day. However, the high turnover relates mainly to the trading of government bonds.

Companies are required to obtain a rating from one of the major credit rating agencies such as Standard & Poor's, Fitch or Moody's prior to listing a bond on the JSE Debt Market.

The SA Reserve Bank reported that South Africa's sovereign risk premium on US dollar- denominated government bonds with a seven-year maturity ranged from 306 to 384 basis points up to August 2023.

A deteriorating economic outlook for resources and for the South African economy in general, as well as the sovereign rating downgrade implied a greater risk for bond investors. The spreads vary in relation to the risk of each bond or note. In 2023, the spread between a government bond and a corporate BBB+ bond was about 173 – 207 basis points.

We will analyse the use of bonds and notes of selected South African companies later in the chapter. We will also look at strategies to determine whether to use fixed or variable rate financing, bank loans or bonds/notes and whether a company should use local or foreign sources of financing.

Why do we expect to see corporate bond or note issues in South Africa? The following are some reasons why companies may use the corporate debt market as a source of debt financing in the future:

- *Interest rates*. Interest rates should be lower on bonds or notes than on bank loans.
- *Credit agency ratings*. The ability of credit rating agencies to evaluate each bond issue and offer a corporate bond a rating is critical for the success of corporate bond and securitisation issues.
- *Low levels of gearing*. South African companies have little debt on their Statements of Financial Position and therefore have the capacity to make debt issues.
- *Use of financial leverage*. This is used to increase the return on equity (ROE) and restructure company balance sheets. A company is able to increase its ROE if the return on assets exceeds the interest rate on bonds. Further, a company's ROE will be further enhanced if the company decides to use the proceeds of the bond issue to undertake a share buy-back (thereby reducing equity).

- *The tax deductibility of interest*. Interest is normally deductible if used in the production of non-exempt income, subject to certain rules.
- *Lower spreads between government bonds and corporate bonds*. The premium to the government bond yields is expected to fall thereby reducing the cost of bond issues.
- *Diversification of sources of debt financing*. This reduces the role of the banks. The consolidation of the banking sector means that companies wish to bypass the banks and issue debt directly to investors.
- *Securitisation and credit enhancements*. Retailers with credit sales can sell their card receivables to a separate entity and then issue debt securities from such an entity thereby separating the risk of the company from its card receivables. The offer of credit enhancements to such securitised issues as well as credit ratings results in a lower cost of financing. Although securitisations were negatively affected by the financial crisis, we Page 13-14 think that there may be demand for quality securitisations in the future.
- *The role of the JSE.* The JSE Debt Market (BESA) has provided an effective and efficient market for the trading of government and corporate bonds (including securitised issues).

We also expect to see the parastatals such as Eskom, ACSA (Airports Company of South Africa) and Transnet listing bonds on the JSE Debt Market to finance infrastructural investments. However, bond issues, particularly for fixed rate bonds, may be constrained until economic conditions improve and interest rates fall.

2 FINANCIAL INSTITUTIONS

The markets cannot operate without the borrowers (those who are seeking funds) and the lenders (those who have funds to invest). The financial institutions bring these parties together by acting as intermediaries for the channeling of funds. While some institutions specialise in certain types of finance, the trend is towards offering a global package of services so that the distinction between types of institutions is becoming blurred.

Banks

Since the services offered by the various types of banks overlap, it is more appropriate to consider the services offered rather than the types of banks.

Commercial banking

Commercial banking includes a wide range of services; the most visible is the current account facility. In addition, a variety of other important services relating to sources of finance are offered. Most important is the overdraft facility, which is a frequently used source of short-term finance in South Africa. Leasing, instalment sale facilities, and factoring are also provided. Commercial banking also includes foreign exchange and providing firms with forward contracts on foreign currency obligations and receipts.

Investment banking

Investment banking, offers a range of facilities and services that includes *inter alia*:

- *corporate finance* – a service offering advice and assistance on mergers, corporate structure, and raising of finance (including debt and equity issues);
- *lending* – the provision of funds, usually of a short- or medium-term nature; and
- *money-market* – actively trading in the money as well as the foreign exchange markets, both for clients and for their own accounts.

One of the most significant impacts that the investment banks have regarding sources of finance is the assistance they provide to companies that need to raise large amounts of finance. Not only can they assist with a public issue of shares, but they can also arrange a private placement for the equity of a company by bringing together the buyer and the seller through their network. Large amounts of loan finance can also be raised through the investment banks, who will act as intermediaries for the borrowers and the investors.

Investment institutions

The term ‘institutions’ usually refers to insurance companies, such as Old Mutual and Sanlam, and to institutions administering pension funds. These bodies are significant participants in the financial markets because of the vast amounts of money under their control.

They are major investors, so a public issue of finance is unlikely to be successful unless backed by one of them. They are therefore often indispensable parties if a

company wishes to raise a large amount of finance, firstly because of the funds they take up, and secondly because of the credibility they bring to an issue. Page 13-15

Private equity and venture capital

Private equity financing is about investing in unlisted companies, although private equity is increasingly used to buy out shareholders and delist under-performing companies. Private equity investors invest capital and management/financial skills into a company. Private equity represents 'hands-on' investing and private equity fund partners become directors in the invested companies. Private equity can be used to invest in new businesses or in turning around struggling or 'tired' companies. The funding is long-term and normally will be for 5 to 10 years.

Private equity financing is often used in leveraged buyouts and management buy-outs, whereby the current management team is financed by the private equity fund to acquire control of the company. Private equity investors achieve high returns and exit their investments by listing on the JSE via an Initial Public Offering, selling to a multi-national company, undertaking a sale to a competitor or by selling their stake back to the management team. Private equity investors achieve high internal rates of return but take on a higher degree of risk. Companies may be initially financed by private equity prior to being listed on the JSE. Warren Buffett is a major private equity investor and takes major stakes in both listed and unlisted companies.

Prior to the financial crisis, the economic environment was favourable towards private equity financing. Why? Interest rates were at reasonably low levels which reduces the cost of a leveraged buyout. South African companies had low debt ratios and refinancing with bonds and senior debt enabled private equity firms to take advantage of interest tax shields. Further, there were South African companies wishing to divest of non-core divisions to focus on international expansion.

Venture capital are equity investments made for the launch, early development, or expansion of a business. When small businesses succeed and grow they need further financing. A venture capitalist provides equity financing for a share of the ownership of a firm. The venture capital markets are less efficient than the public markets as transaction costs are far higher and there is a small number of buyers and sellers. There are no reporting requirements, and information is not freely available to all parties.

In the late 1990s, venture capital financing of IT companies was very popular, but the collapse of many IT companies in 2001 reduced the funds available to IT start-ups.

In the last five years, however, there has been a resurgence of venture capital financing for IT firms with cloudbased and online business models. Another sector that attracts venture capital funding is the biotech sector. In 2014, venture capital funds were again generating high returns. Facebook and Twitter were financed by venture capital funds prior to listing. In South Africa in 2022, funds raised by private equity amounted to R19.6 billion and R15.2 billion was invested by private equity funds in 2022. It is reported by SAVCA in 2023 that the private equity sector has R213.3 billion in funds under management.

Why would a company use private equity financing? Firstly, it may be the only financing available to the company. The banks may not be prepared to lend to the company. Secondly, the company acquires the financial and management expertise of the private equity partners who provide strategic, operational and financial advice. This is important in relation to raising debt and equity finance and in structuring the financing of the company, as well as optimising the exit price, either in the form of an IPO or a trade sale. Thirdly, private equity investors often have an extensive business network, which is important for the expansion of the company's operations and sales as well as the recruitment of key people. Fourthly, the private equity investor provides long term equity capital which offers the company stability as well stepping in at later stages with further funds if the results merit further expansion.

Edcon was acquired by a private equity firm, Bain Capital for over R25 billion financed with bonds and equity and the group was delisted from the JSE[1]. Management was involved in Page 13-16 the buyout as it is important that management is properly incentivised to generate the future returns required by the private equity firms. Hedge Funds have also been active in financing private equity deals. Even football clubs have been affected, for example Manchester United which had been listed was taken over by private investors in a highly leveraged structure.

What do private equity firms look for in an investment? Whilst not all private equity firms are the same, the following may reflect some of the requirements for investing:

- Experienced superior management in place and a strong financial team.

  Sometimes, the private equity firm will replace current management if the firm has been earning poor returns.

- Significant equity interest held by management in the firm.

- An attractive purchase price. This may be the most important factor in driving future returns. It is important to be disciplined particularly when there is bidding by a number of private equity firms.
- Sustainable competitive advantage in the firm's key markets.
- Growth potential and opportunities to expand into other markets.
- Sustainable cash flow from operations after taking into account working capital and capital expenditure requirements.
- Undervalued assets and availability of underutilised property assets.
- Low debt-equity ratios.
- Available exit strategies.

Invenfin, which is part of Remgro, invests in innovative firms with patents that can be protected and which creates a barrier to entry. There must be a global potential for the product. Innovators should be committed and passionate but must be open to Invenfin's guidance.

Due diligence is an important component of any private equity investment. This involves an extensive analysis and audit of the financial numbers and an evaluation of the products and financial projections. It is important to verify the existence and value of assets and liabilities as well as evaluating the people and the firm's business model. We will come back to due diligence in Chapter 17.

Private equity (PE) firms obtain funds from investors and then search for companies to invest in, take an active role on the board of directors and work with those companies to grow operations and achieve high returns over a 5-10 year period. Private equity funds are often structured in the form of limited private partnerships with a stated investment period of 10 years. The investors are limited partners and often will consist of pension[2] and retirement funds, financial institutions and insurance companies. The manager of the fund will be the general partner and together with the limited partners will own the private equity fund.

Who are the major private equity participants in South Africa? Ethos, Brait, IDC, Sanlam, Investec and Invenfin are the main participants in South Africa but increasingly there are international private equity firms that are looking to invest in South African companies. Ethos (now part of the Rohatyn Group) is a leading private equity firm in South Africa and has a carrying value of invested capital of R2.7bn

which is invested in 22 portfolio companies. We will now evaluate a typical private equity transaction. In 1998 Ethos, purchased a majority stake in the JSE listed company, BTR Dunlop for R2.05 per share. In 2002 Ethos purchased the remaining 42% for R4.05 per share and delisted the company. Ethos, with management, then undertook the disposal of non-core assets and divisions, reducing debt levels and returning capital to shareholders. Ethos organised financing made up of senior debt, mezzanine finance, and working capital and export debtor financing totalling R670m. In 2006, Ethos exited the investment via a trade sale to Apollo, a major listed Indian tyre manufacturer.

Another example relates to Ethos' 2006 investment in Holdsport in a R681m buyout. Holdsport is a sporting, leisure and recreational goods retailer and includes Sportsman's Page 13-17 Warehouse. The number of stores increased from 35 to 51 stores. Holdsport achieved an annual EBITDA growth rate of about 15% per year under the guidance of Ethos, which exited its investment through a successful listing (IPO) on the JSE at an enterprise value of about R1.5bn in late 2011. In 2014, the market capitalisation of Holdsport was around R1.85bn. which increased by about another 50% prior to being acquired by Long4Life in October 2017.

Ethos reported in 2018 that it had exited 95 investments. Ethos had generated a 25.2% gross realised IRR and a 2.8× multiple of invested capital (MOIC). Refinancing operations with loans can enhance returns, and we will come back to explain this in Chapter 17. Other private equity transactions undertaken by Ethos include the following investments:

Investment	Year	Size (Rm)	Investment	Year	Size (Rm)
RTT	2014	Not disclosed	Tiger Automotive	2008	1032
WACO International	2012	Not disclosed	House of Busby	2008	1319
Kevro	2012	850	Brand∞rp	2007	1320
Universal	2011	1300	Alexander Forbes	2007	8700
Idwala	2009	2900	Eaton Towers	2015	USD150m
Eazi Access	2016	1600	Channel Vas	2018	700

Due to the higher risks of private equity and the active role that the private equity team plays in each company, private equity firms will often require an investment to promise an IRR in the range of 20% to 30%, although this is often dependent on financing structures and the size and nature of the firm. Private equity firms also play a crucial role in enabling BEE participation in companies. Private equity may involve delisting a company – a public-to-private transaction – and then restructuring the firm

and perhaps relisting or engaging in a trade sale at a later stage. This is what Ethos did with Alexander Forbes and Brandcorp. Previously, Brait undertook a public to private transaction with the delisting of Pepkor.

Ethos' focus is on investing in medium to large companies which have strong management teams and meaningful equity participation by management. Firms should offer good quality growth opportunities and should offer 'clear competitive advantages and have critical mass to sustain the business through economic cycles'. Investments are mainly in the R750m to R3bn range.

If the transaction involves high levels of debt, then private equity firms will often invest in companies which offer reasonably stable operating cash flows, which do not need large amounts of capital expenditure in the future and which may offer opportunities to divest of non-core divisions. Brandcorp issued high yield notes in late 2010 and this was connected with ABSA's financing of the leveraged buyout by Ethos in 2007. RMB Ventures achieved a successful private equity exit with the listing of Life Healthcare in 2010. It was reported that RMB Ventures achieved a net profit of R800m from this listing. In 2023, Life Healthcare had a market capitalisation of R30bn. It listed at a market capitalisation of R14bn. Alexander Forbes made a return to the JSE in 2014 with a market capitalisation of about R10bn but which has since fallen to R7.89bn in October 2023.

The SAVCA 2023 survey reported that the private equity sector in South Africa had R213.3bn of funds under management and this included undrawn commitments of R48.6bn. The average deal size was R80m. There were 72 new investments and 117 follow-on investments. According to the SAVCA survey, 7% of private equity investments relate to start-up and early stage investments while 9% are buyouts, 16% relates to replacement capital and 68% of unrealised investments relate to expansion and development. The proportion of proceeds to investment was only 1.6 in 2022 as compared to 2.7 in 2020. Exits to trade buyers generated the highest proceeds to cost of 2.7 times. One needs to question whether this represented an adequate return when we consider relative risk profiles, liquidity and potential leverage/ refinancing effects. Ethos reports a record of delivering an IRR of 24% over 39 years prior to its merger. In 2023, Ethos (TRG) explained that it remained focused on sectors subject to structural tailwinds. The key themes are as follows: Page 13-18

- Growing connectivity & data access
- High impact differentiated services.

- Demand-led growth in consumer goods
- Technology/AI driven disruption to “business as usual”
- Transportable business models scaling across countries.
- Lowest cost-to-serve, scalable businesses.

The relevant sectors are telecommunication services, technology enablers, MedTech, ICT, Financial Services, FinTech, Specialist/niche retail, FMCG and Business/consumer services.

Special institutions

Some institutions are created, either formally or informally, for specific purposes. Examples include the Industrial Development Corporation (IDC) and Business Partners.

Industrial Development Corporation

The prime consideration of the IDC in respect of any application is its economic merit, which it defines as the ability of the business to make acceptable profits within a reasonable period. The IDC prefers that its exposure should not exceed that of the owners. The IDC is therefore normally prepared to consider loan finance, which is approximately equal to the owners’ interest in the business. Although the emphasis is on development (the establishment of new industries or the expansion and modernisation of existing industries), the financing of buy-ins or takeovers with significant entrepreneurial development could also be considered should the transaction be empowerment related (BEE). Job creation is also an important factor. For BEE transactions, the IDC may reduce the investment requirements by the owners. Acquisitions need to be by either a historically disadvantaged person or majority black-owned business and at least 50% of the funding needs to be reinvested for growth and expansion. Evidence of due diligence as well as a DCF valuation is required to confirm the company value.

Preference is given to:

- financing fixed assets and the fixed portion of growth in working capital requirements;
- green field projects, expansions and rehabilitations;

- relatively sizeable projects;
- projects which have economic merit in terms of profitability and sustainability; and
- projects which have developmental impact in terms of jobs, exports, spatial, black economic empowerment and downstream opportunities.

The IDC normally requires security, compliance with environmental requirements and a reasonable financial contribution by the owners.

The replacement of existing sources of finance falls outside the scope of the IDC's activities. The IDC will take an equity stake in a company and SASOL is a prominent example of an IDC supported investment. The IDC also invested in the Mozal smelter with BHP Billiton. Increasingly, the IDC is focused on the development of small and medium enterprises, the financing of BEE entities and lending across the African continent. Examples of IDC financing include the funding of BEE investments in Aspen Pharmaceuticals, Afrox Healthcare and KWV. Usually the minimum loan amount is R1 million and the IDC will take an equity stake. The interest rate may be linked to the prime lending rate and any equity investment by the IDC in the past was required to provide a real after-tax IRR of 8%.

The IDC also provides financing for development initiatives with interest charged at prime less 2% or 3%.

The IDC has a number of strategic business units, including a venture capital unit, which provides funding to start-up technology-focused firms that plan to develop products and bring Page 13-19 them to market. The IDC will provide finance of between R1m and R5m per transaction and in the past required a minimum before-tax IRR of 3% "plus upside". The IDC will also fund green industries, agricultural entities and other traditional sectors.

Public Investment Corporation (PIC)

The PIC is an investment management company whose largest client is the Government Employees Pension Fund. The PIC has assets under management of over R2.5 trillion. Whilst the PIC mainly invests in fixed income securities and listed companies, the PIC is increasingly playing an important role in financing BEE transactions. In 1999, the Isibaya Fund was created to invest in socially responsible projects which will result in job creation, wealth redistribution, poverty alleviation and

infrastructural development. The focus is on transformation and maximising financial returns. The fund invests between R100m and R500m per transaction for five to fifteen years. The fund uses the risk free rate as the base required return and adds a risk premium depending on the risk implicit in the investment. The fund wishes to see active BEE management and ownership in companies that it invests in. Other important aspects are an attractive entry price, the financing of new and broad-based BEE participants, skills transfer and significant BEE ownership when the PIC exits the investment.

The fund wishes to co-invest with other financial institutions and projects are required to have positive environmental, societal and governance impacts.

National Empowerment Fund (NEF)

The NEF falls under the auspices of the Department of Trade and Industry (DTI). The NEF will finance transactions that will result in meaningful BEE ownership and BEE participation in the operations of the company. The NEF provides a blend of loan finance and equity financing to BEE companies and endeavours to reduce the cost of capital facing BEE firms. The BEE investment should be sustainable. The NEF will warehouse shares (temporary purchase and later on-selling to BEE entities) and provide a secondary market for the buying and selling of BEE stakes in companies. The Fund can also underwrite initial public offerings by BEE companies listing on the JSE or Altx. The NEF also acts as a strategic advisor to companies making Black Public Offers such as SASOL. The CEO of the NEF, Ms Philisiwe Mthethwa, further emphasised the role of the Strategic Projects Fund through which the NEF provides funding for broad based BEE participation in early stage projects in sectors such as business process outsourcing, mining, renewable energy and agro-processing and tourism. Regional economic development is important.

Small Enterprise Finance Agency (Sefa)

Sefa provides financial products and services to SMMEs and Co-operatives which includes direct lending as well as credit guarantees. Sefa is a subsidiary of the IDC. SMMEs and Cooperatives can obtain credit facilities from R500 to R5m. Khula Enterprise Finance, which falls under Sefa, is focused on assisting small and medium sized enterprises (SMEs) to access capital from banks, agencies and financial institutions. For example, a small business will require tangible collateral (such as a

covering bond over a residential or business property) to qualify for a business loan from FNB. Yet FNB also states that 'Khula guarantees are available to clients who have a well-run business but lack collateral to secure a loan'. FNB will lend on the basis of Khula (Sefa) credit guarantees.

Business Partners

Business Partners was formed to promote entrepreneurship among small and medium enterprises. It will invest capital between R500 000 and R50m depending on the owners' contribution, risk and collateral, as well as provide guidance. The company will invest in startups, expansions, management buy-outs and buy-ins, and franchises. Business Partners will often Page 13-20 use a combination of financing structures including equity, term loans, royalties, depending on the security provided and the business risk of the investment. Often, there will be a loan and Business Partners will also take an equity interest in the company. The repayment of any loan will be over a period of 5 to 10 years and interest rates are linked to the prime lending rate. Companies receive practical advice and mentoring from Business Partners' managers. There is also a focus on providing finance to black economic empowerment (BEE) entities. Business Partners is increasingly combining private equity investing with the provision of loan facilities. Business Partners provides funding to businesses with gross assets of less than R100m and sales revenue below R200m.

3 EQUITY-RELATED INSTRUMENTS

The primary source of finance is that provided by the owners of a business. In the case of a company this is the share capital and the retained earnings, and is referred to as the equity. The equity may take various forms.

Ordinary shares

The interest of the ordinary shareholders is directly related to the performance of the company. Their wealth increases by the capital growth in the shares and the dividends received. On the other hand, when the company is performing poorly, this may be reflected in a falling share price. The return to the shareholder (dividends plus capital growth) is on average higher than could be received on interest-bearing investments. This is because the investor is bearing greater risk in holding an equity investment.

The company can raise finance by the issue of new shares which the public may take up. While shareholders include the general public, most shareholders or investors

are the pension funds and insurance companies, as well as other large corporations. Provided the issue price is acceptable, there will be no shortage of willing buyers for the shares. Issue expenses have to be met from the finance raised.

Share-issue expenses usually include the following:

- *Listing fee* – depends on the monetary value of the securities listed;
- *Advertising expenses* – these would include the cost of drawing up and distributing brochures and prospectuses;
- *Professional fees* – legal, audit, sponsoring stock broker, transfer security and JSE fees for drawing up the prospectus and for other related professional services;
- *Banker's fee* – the issue is usually administered by an investment bank, the fee for which may be 1-2% of the gross proceeds of the issue;
- *Underwriting commission* – the issue is usually underwritten to protect against the issue not being fully subscribed. A commission of about 2-4% of the amount underwritten is usually charged, but this could increase depending on the risk of the issue not being fully subscribed; and
- *Taxes and duties*.

Retained earnings

Part of the ordinary shareholders' equity consists of earnings which are available for distribution but which have been retained in the company. If they are distributed as dividends, replacement equity funds would eventually have to be raised. Hence the retention of earnings constitutes an immediate source of finance. While it is a simple matter to retain earnings as a source of funds, this is not without cost. The shareholders have, in effect, agreed to forego the dividend in the expectation of future growth, and these expectations have to be met. The cost of retained earnings is therefore the same as the cost of new equity, except that there are no issue expenses. The decision to retain earnings rather than pay a dividend is determined by the dividend policy of the company.

Preference shares

Preference shares are a hybrid, having some of the characteristics of equity and some of debt. Typically, preference shares promise a fixed dividend and in this way preference shares may be likened to debt. However, if the company has insufficient

funds to pay the preference dividend, this dividend may be passed. In this way they may be likened to equity in that the company does not have a legal commitment to pay a dividend. The nominal rate for preference dividends is usually based on the after-tax rate for long-term debt. This is because while the interest on the debt is deductible for purposes of taxation, the preference dividend is not.

Unless the preference shares are specifically designated as non-cumulative, they are assumed to be cumulative preference shares, and the preference shareholders have a right to arrear dividends. The arrear dividends accumulate until they are paid. In the event of the preference dividends being in arrears, the preference shareholders are granted voting rights and thus can exercise a measure of control over the company.

Dividends paid by the company, whether they are ordinary dividends or preference dividends, will attract dividend tax. As this is a withholding tax, it is a tax for the shareholders and not the company. Also, dividends payable to South African resident companies are exempt from the dividend withholding tax. The current dividend withholding tax rate is 20%.

Example 13.1:

Siyanda Ltd has entered into discussions with InvestBank Ltd about raising R10 million to acquire 51% of the shares in another company. For a company of Siyanda's risk class the market rate of debt is 9%; alternatively, preference shares may be issued at 74% of the prime lending rate, which is currently at 10%. Siyanda will not be able to deduct interest on the loan as it will be used to acquire shares rather than operating assets.[3]

We will assume that a dividend withholding tax of 20% applies. As this is a tax due by shareholders, we will not include this tax in our analysis. InvestBank is exempt from paying tax on dividends received from a resident South African company. The position from Siyanda's perspective when comparing preference shares and debt is as follows:

Siyanda	Prefs	Debt
Interest paid	-	900,000
Dividend paid	740,000	
Tax shield	-	-
Cost to Siyanda	740,000	900,000

The position from InvestBank's perspective is as follows:

IiivestBaiik	Prefs	Debt
Interest received		900,000
Dividend received	740,000	
Tax paid	-	-243,000
Tax on dividend - exempt	-	
Net income to InvestBank	740,000	657,000

We can see that where the company is not able to obtain the tax shield from borrowing, then using preference share funding will minimise the company's after-tax cost of financing. If the interest on the loan had been deductible, then the net cost to Siyanda would have been R657 000 and there would have been equivalence in the after-tax position for Siyanda and InvestBank. Page 13-22

Participating preference shares

These are preference shares that have elements of both preference shares and ordinary shares. They have the fixed dividend, as do other preference shares, and they share in the remaining profits of the company in some predetermined proportion to the ordinary shares.

Redeemable preference shares

These are the same as the normal preference shares except that the company is required to redeem them at a specified price on a particular date or over a given period of time. They are attractive to the company because they offer a degree of flexibility. Firms will generally issue redeemable preference shares but the term is required to be longer than 3 years to ensure that such preference share funding is not classified as debt in terms of tax legislation.[4]

Convertible preference shares

These are similar to normal preference shares except that the holder has the right to exchange them for ordinary shares or other securities according to prearranged terms. If the holder believes that the ordinary shares (or other securities) offer a better risk-return relationship than the convertible preference shares, conversion will take place; if not, he or she will elect to retain the preference shares and forgo conversion.

Why do firms issue preference shares? The issue of preference shares will enhance the Statement of Financial Position of a company as the amount is recorded under equity capital and preference shareholders cannot place the company into bankruptcy.

This will reduce the financial leverage of the company without causing dilution for the ordinary shareholders. For companies unable to use the tax shields from debt, the issue of preference shares is a better option than debt. Further, the issue of preference shares will not affect the control of the company.

The preference dividend rate will be issued at a rate of about 70–75% of the prime lending rate. This means that although the company does not receive a tax shield, the dividend rate effectively reflects the after-tax cost of debt, although large corporates are able to borrow at rates below the prime lending rate. The issue of preference shares is useful in BEE transactions as the company may be an investment company acquiring shares in an operating company and therefore interest would not be deductible.

Interest is deductible only in the production of taxable income and the dividends accruing from the shares are not taxable. However, it is important to consider whether section 240 applies.

4 DEBT INSTRUMENTS

The distinction is often drawn between long-term and medium-term finance, although the exact line of separation is hazy. Normally medium-term finance would be for periods of not less than one to three years and up to between five and ten years, whereas long-term finance would be for longer periods.

There are some other classifications of debt which may be useful, one of which is according to variability of the interest rate on the debt:

- *Fixed interest*. This is debt that bears a fixed rate of interest for the period of the debt issue. Many debentures (bonds) fall into this category.
- *Variable interest*. In this case the interest is not fixed for the duration of the debt but fluctuates according to a benchmark rate such as JIBAR or the prime overdraft rate. Mortgage loans are a common example of variable-interest debt.

Another common classification relates to the security on the debt:

- *Secured debt.* This is debt that is secured over one or more of the assets of the company. In the event of the liquidation of the company, the proceeds from those assets would first be applied to satisfying the claim of the secured creditor.[5]
- *Unsecured debt*. This is debt that does not have preferential claim over any of the assets of the company. It is at greater risk than secured debt, so it should command

a higher interest rate, all else being equal.

Some of the forms of debt available in South Africa are described below.

Corporate bonds, notes and debentures

We will use the terms debentures, notes and corporate bonds interchangeably. A debenture arises out of a contract between the company issuing the debenture and the investors. The terms and conditions of the contract are contained in the debenture trust deed which is a contract between the company and the trustees. The trustees are the representatives of the debenture holders and it is their duty to ensure that the rights of the debenture holders are observed.

Listed debt securities on the JSE are often called notes or corporate bonds. Notes will often represent medium-term debt securities with terms of three to five years and are more likely to be issued at variable interest rates. Corporate bonds and notes are often issued in denominations of R1 million per note/bond.

The main terms of any bond or note issue are the interest rate and repayment conditions. Debentures and bonds are usually issued at a fixed-interest rate, and secured over certain assets, but neither of these conditions is necessary.

Bond, note and debenture issues often include restrictive covenants, which may limit the freedom of the management in running the company. Covenants are inserted to protect investors as covenants usually restrict the risk to which the company is exposed.

Typical restrictive covenants may be designed to:

- ensure that certain financial ratios remain within agreed limits, e.g. debt ratio, quick ratio;
- restrict the raising of further loans; and
- restrict the payment of dividends.

Debentures, notes and bonds may have differing terms and conditions regarding security, period, and yield. While every possible variation will not be discussed, the following are included:

- *Secured*. The debentures/bonds/notes are secured over certain assets of the company or there is a floating charge over all the assets of the company.

- *Convertible*. The debentures/bonds/notes are convertible into other securities, usually shares, on prearranged terms.
- *Redeemable*. The debentures/bonds/notes are redeemable at the option of the company before maturity, or on specified dates, according to the terms of the issue.
- *Fixed rate*. The interest rate is fixed for the duration of the bonds/debentures.
- *Floating rate (variable rate):* The interest rate is a floating rate that varies with a reference rate such as ZARONIA or JIBAR. For example, the floating rate may be 3% (300 basis points) above ZARONIA or JIBAR.
- *Specified denomination (principal amount):* The interest payable per bond/note will be based on the principal amount of each bond or note. In many cases, the specified denomination is R1m per note/bond. Page 13-24
- *Participating.* The debenture holders receive a fixed proportion of the profits of the company in addition to their interest entitlement.
- *Income*. Interest is payable only if the company has sufficient earnings to meet this commitment.

Secured debentures or bonds may mean either that the debentures are secured in terms of specific assets or that there is a floating charge over all the assets of the company except for assets that have been offered as security for specific loans. Companies prefer a floating charge over their assets as this offers flexibility in relation to the management of the company's assets. If a company defaults on interest or capital payments, then the debenture or bond investors are able to force the sale of the assets that have been offered as security.

Companies often issue debentures and corporate bonds at fixed interest rates until maturity. However, companies may issue bonds that offer a fixed interest rate for a period such as 5 years, at which time the interest rate becomes a variable rate, which will be at a margin to a market rate such as the bank swap rate, ZARONIA (South African Overnight Index Average) or JIBAR (Johannesburg Inter-bank Agreed Rate) or the bank bill rate. Notes are often issued at a variable interest rate based on a reference rate such as JIBAR.

Debentures and bonds may include a call option. The bonds may be redeemable in 10 years' time, but the company has the right to purchase the bonds after a period such

as 5 years. This means that if interest rates have fallen, the company is able to call in the bonds and reissue the bonds at the lower market rate.

Although we will use the terms debentures and bonds interchangeably, debentures tend to be secured, whilst bonds are usually not secured. Corporate bonds and notes may be rated by a credit rating agency such as Standard & Poor's, and most corporate bonds and notes in South Africa are issued by large listed companies with strong credit ratings. Corporate bonds and notes also refer to issues by companies whilst debentures are now issued mainly by finance companies. A company may issue bonds, notes or debentures via a private placement with financial institutions or the company may make a public issue. Any issue of corporate bonds or notes will fall under the Financial Markets Act. Bonds may be issued on the JSE Debt Market, which will require that the company submits a disclosure document indicating the terms of the issue, financial information, the purpose of the issue and information on the shareholders and directors. The company is also generally required to obtain a rating from a credit rating agency such as Fitch or Standard & Poor's. This represents an independent evaluation of the credit risk of such an issue. In any case, a company will obtain a credit rating to ensure the success of the bond issue with investors and to minimise the interest yield in relation to the government bond yield. The difference between the bond yield and the government bond yield is often termed as the bond's spread and is stated in terms of basis points (100 basis points = 1%).

Although South African companies are making increasing use of corporate bonds and notes rather than bank loans, the issuing of bonds is limited to the larger companies such as Barloworld and MTN. The amount of bank loan financing far exceeds the financing from corporate bonds or notes. Investors in corporate bonds are not offered specific security and rely on the creditworthiness of the firm as a whole. Investors in corporate bonds will place reliance upon the firm's rating by rating agencies such as Standard & Poor's, Fitch IBCA and Moody's. For example, the rating of BHP by Standard & Poor's is "A+", which indicates a strong credit rating for BHP bonds. BHP has committed itself to a regular bond issuance program and will be making less use of bank loans. The issue of bonds directly to the capital markets has been driven by demand factors and low interest rates. Companies are more willing to issue bonds or debentures at a fixed interest rate due to the fact that interest rates are at historically low rates. **Page 13-25**

The effect of issuing corporate bonds is to reduce the cost of debt financing to companies, thereby reducing a company's cost of capital.

Long-term loans

Mortgage loans

A mortgage loan is a long-term loan which is usually secured over the fixed property of the company and which bears a variable rate of interest.

Mortgage loans can be raised against the value of the property offered as a security. The amount that can be raised depends primarily on the value of the property and the purpose for which the loan is raised. However, various institutions may offer differing amounts on the same property. It should be possible to raise a mortgage bond amounting to 70–80% of the valuation of the property for industrial purposes.

Bank loans

The major source of long-term financing consists of bank loans, particularly for medium and smaller companies. Loan finance may take the form of a term loan, which is a loan that is repayable over a fixed period such as five years and will often relate to a specific financing requirement, such as the investment in plant and equipment. Otherwise, a bank loan may be structured in the form of a loan facility which the company draws down as required. The loan will be subject to an interest charge and the company is also required to pay for the right to use the loan facility. The loan financing decision requires that management decide on a number of key issues:

- secured or unsecured loan;
- variable rate or fixed rate loan;
- period of loan and timing of capital repayment; and
- loan or draw-down facility.

Whether the company is required to offer security, may be beyond management's prerogative. The bank may demand security. The provision of security to the bank will result in a lower interest rate but reduces the flexibility of management to manage the company's assets. An increasingly important decision relates to whether the company should take on variable rate or fixed rate debt finance. We will analyse this separately. The period of the loan will be driven by the nature of the financing requirements. For example, assume that the company wishes to invest R20 million in new plant and equipment with an economic life of 10 years. The payback of this project is expected

to be 5 years. In making the financing decision, management will organise a long-term loan with a term of at least 5 years.

If the project's cash flows are expected to occur evenly throughout the life of the project, then management may wish to repay the loan in equal instalments over the period of the loan. If the project cash flows are weighted towards the end of the payback period of the project, i.e. 5 years, then the company may wish to repay the loan amount as a *bullet* or *balloon* payment at the end of 5 years. A firm will wish to structure some flexibility into the loan repayment just in case project cash flows take longer to be realised than expected in terms of the projections. In our case, it may be prudent for the company to enter into a 5-year loan but also to organise a further loan facility in case project cash flows take longer than expected. A loan facility offers the company greater flexibility to use loan finance only when required in terms of the company's operations. If a company generates strong cash flows then the company will use only part of the facility. Otherwise, a straight bank loan may result in the company having cash resources it does not require. Page 13-26

Fixed interest rate loans

Bank loans can be set for a fixed period of time, normally up to 5 years and at a fixed interest rate. At the end of the period, the loan may be renewed and the interest rate may be reset to the market rate at that time. The repayment of a loan may be structured in the form of equal monthly, quarterly or annual repayments or the loan may require the company to make only interest payments throughout its term and the repayment of the capital amount will occur as a single payment at the end of the period of the loan.

For example, ROC Ltd wishes to expand its distribution facilities at a cost of R8m. The company is considering a bank loan of R8m at an interest rate of 11% per year, which is repayable in equal annual instalments over 5 years. Alternatively, the bank is also prepared to offer the company a loan which is repayable as a bullet payment at the end of 5 years. What are the effects of either loan option on the company's financing cash flows?

Firstly, the repayment of the loan in equal instalments will result in the following annual repayments, which include interest and repayment of capital:

Year	Beg.Bal	Payment	Interest	Principal	End. Bal.
1	8,000,000	2,164,562	880,000	1,284,562	6,715,438
2	6,715,438	2,164,562	738,698	1,425,864	5,289,573

Year	Beg.Bal	Payment	Interest	Principal	End. Bal.
3	5,289,573	2,164,562	581,853	1,582,709	3,706,864
4	3,706,864	2,164,562	407,755	1,756,807	1,950,056
5	1,950,056	2,164,562	214,506	1,950,056	0
Total		10,822,812	2,822,812	8,000,000	

Use a financial calculator to determine the annual payment, as follows:

N	I/YR	PV	PMT	FV
5	11	-8,000,000	**2,164,562**	0

If the company decides to take a loan that is repayable at the end of the term of the loan then the annual interest payments will amount to R880 000 [R8m × 11%] per year and the company will be required to make a bullet payment of R8m at the end of 5 years. The financing cash flows relating to either option are stated as follows:

Cash Flows	0	1	2	3	4	5
Loan with equal instalments		-2,164,562	-2,164,562	-2,164,562	-2,164,562	-2,164,562
or						
Loan with capital payable at end of term		-880,000	-880,000	-880,000	-880,000	-8,880,000

Which loan option should the company select? The answer depends on a number of factors. Firstly, if the project cash flows are expected to occur evenly over the period, then a loan that is repayable in equal instalments may be preferable. If the project is expected to generate variable cash flows, then the company may prefer the single repayment at the end of 5 years. If the project is expected to have a significant market value at the end of 5 years, then a single repayment may better match the cash flows from the project, particularly in relation to the cash flow from the sale of the asset. A single repayment does increase the risk to the company and the bank. Perhaps owing to weak cyclical economic conditions, the company is not able to repay the full amount at the time or is not able to organise refinancing.

Assume in our example that the project is expected to generate operating cash flow savings of R4m per year over 5 years. This means that the company is able to meet the financing payment of either financing option. However, assume that there is a 20% chance that the savings will occur as R1m in Year 1 and R7m in Year 2. Although the project remains positive, the company will not be able to make the annual instalment due in Year 1 and the bank could call in the loan and place pressure on the firm to dispose of its assets to repay the loan.

An advantage of a fixed rate loan is that the company knows the interest rate it is required to pay over the next 5 years. If interest rates rise over the term of the loan, the

company's financing and interest cash flows are not affected by the change in monetary conditions.

Variable interest rate loans

A variable rate loan means that the company will be charged an interest rate that will change as general interest rates change. Often the interest rate charged will be at a premium to a market rate such as the 90-day bank bill rate, the ZARONIA rate or the JIBAR rate. For offshore loan financing, interest rates will often be issued at a premium to SOFR (Secured Overnight Financing Rate) or SONIA (Sterling Overnight Index Average).

Usually interest is determined on a daily basis and is linked to a base rate, and a bank can structure the loan repayments to meet the cash flow requirements of the firm. A firm will try to match repayments to the cash flows and the term of the underlying project. Usually, banks will offer firms a redrawing facility if the firm pays any capital amount early. A fully drawn advance type loan offers firms flexibility in terms of repayment terms. Firms can also borrow on a *capped rate* basis, usually for up to 3 years. This means that the company will pay a variable rate but the interest rate cannot increase above a specified rate. However, the company will benefit from any fall in interest rates. For example, a company may borrow at a variable rate of 7%, capped at 10%. If interest rates rise to 9%, the company pays 9%. If interest rates rise to 12%, the company pays the capped rate of 10%. Increasingly, companies make use of interest rate swaps to either move from fixed to variable rate financing or to move from variable to fixed rate financing. We will explain interest rate swaps and caps in greater detail in Chapter 18.

Loans may be subject to restrictive covenants. What are some restrictive covenants that we see in practice?

- *Working capital*. The firm may be required to maintain a specified current ratio.
- *Restriction on further borrowings*. The firm may not be able to undertake further borrowings without the permission of the bank.
- *Restrictions on asset sales*. The firm may require the bank's permission prior to making any disposals of major assets. This is particularly true of sale and leaseback transactions.
- *Restrictions on lease financing*.

- *Provision of regular financial statements and cash flow budgets* and maintenance of financial ratios.
- *Restrictions on dividend payments.*

Unsecured loans subject to a negative pledge

Companies can borrow on an unsecured basis but may offer a negative pledge. What does this mean? In terms of the negative pledge, the company agrees that it will not offer security or pledge assets to other lenders without the prior consent of the current debtholders. A negative pledge protects the current debtholders' rights to the company's assets without affecting management's flexibility to manage the company's assets.

Unsecured notes

Companies can issue notes which are similar to corporate bonds. These are usually redeemable but may be perpetual, and are not secured. In fact, often many issues will be subordinated and will rank just above preference shares and ordinary equity but behind most other creditors. This enables companies to obtain the advantage of the interest tax shields and not let these issues impact negatively on the company's rating when it comes to senior Page 13-28 debt financing. Unsecured notes will offer a higher interest rate than secured debentures. Perpetuals do not have a maturity date and investors have to sell in the secondary market to realise their capital. The company does not have to raise finance to repay the capital amount at maturity.

Foreign loan or bond financing

Firms use foreign markets to raise debt finance or to issue corporate bonds. Companies such as BHP, who will earn US dollar revenues from the sale of resources effectively hedge by borrowing in US dollars. For example, BHP states in its 2017 annual report that the group's policy is "to maintain borrowings and excess cash predominantly in US dollars". Companies may borrow in US dollars and Japanese yen and hedge the currency exposure with either forward contracts, futures or other derivatives. What are the advantages of borrowing from offshore sources? Firstly, the company may have foreign currency earnings or operations. Secondly, interest rates may be lower, even after hedging the foreign currency exposure. Thirdly, offshore

centres such as London may have deeper and more liquid markets, which enables the firm to issue debt at a lower interest rate.

Credit ratings

The rating agencies play a pivotal role in the corporate bond market, particularly when firms wish to bypass the banks and obtain finance directly from investors and pension funds. A bank normally performs two important functions when it comes to providing loan finance to companies. Firstly, a bank will have the funds to lend and secondly the bank assesses the risk of a borrower in terms of repaying capital and interest. Loan managers at banks assess credit and business risk and continue to assess the borrower's ability to pay, normally on an annual basis. A company can issue bonds directly to the public or to pension funds. However, the secondary role of risk assessment is still required and this is where the credit rating agencies play a critical and independent role. They rank the company's debt and offer a score indicating the risk of non-repayment of capital and interest. Credit rating agencies assess the risk of short-term and long-term debt issues. The ratings are used to determine the interest rate that companies will be required to offer on debt issues. Who are the rating agencies? There are three main rating agencies, Fitch IBCA, Standard & Poor's (S&P) and Moody's, who produce opinions on the creditworthiness of specific debt issues. They have similar rating systems and we have focused on S&P. The long-term debt ratings and their meanings are set out in Table 13.2.

Table 13.2 Ratings by S & P and Moody's of long-term debt issues.

S&P	Standard & Poor's - meaning	Moody's	Meaning
AAA	The capacity of the company to meet its financial commitment on the obligation is extremely strong.	Aaa	Extremely strong and highest credit quality
AA	The capacity of the company to meet its financial commitment on the obligation is very strong.	Aa	Very strong credit quality
A	The capacity of the company to meet its financial commitment on the obligation is strong. The company is somewhat more susceptible to the adverse effects of changes in circumstances and economic conditions.	A	Strong, high credit quality
BBB	An obligation rated BBB offers adequate protection parameters. Adverse economic conditions are more likely to lead to a weakened capacity of the company to meet its financial commitment on the obligation.	Baa	Good, adequate credit quality
Non-investment grade			
BB	Obligations rated BB to C are speculative	Ba	Somewhat Speculative
B		B	Speculative
CCC		Caa	Risk of default
CC		Ca	High risk of default
C		C	In default with interest

Reference: Standard & Poor's Corporate Ratings Criteria, 2002

What do the ratings mean in terms of risk of default? A debt issue rated as **AAA** is likely to have about a 0.5% chance of default within 10 years, whilst a debt issue rated as **A** is likely to have a 1.2% chance of default within 10 years. A debt issue rated **BBB** is likely to have a 2.9% chance of default within 10 years whilst there is an 11.8% chance that an issue rated as **BB** will default within 10 years.

The facts are that South African investors have been much more conservative and do not normally like to buy bonds rated as BBB and below. In the USA, about 25% of bond investors do invest in below investment grade bonds.

What do credit ratings mean in terms of interest rates charged to firms? In the USA on 2 October 2023, Aaa 10-year bonds were trading at a spread of about 77 basis points above the government bond yield, whilst Baa bonds were priced at about 178 basis points above the government bond yield.

In South Africa, corporate bonds of large companies were trading at spreads of about 1.44% (144 basis points) above comparable government bond yields in late 2018. Determining the spread is important for the valuation of bonds. For example, if the current government bond yield is 10.56%, we would add 1.44% and discount these corporate bonds at a rate of 12% per year.

In 2014, Standard & Poor's downgraded the sovereign credit rating of South Africa from BBB to BBB-, which is one grade above junk status. This downgrade increased the cost of funding for the South African government and South African companies. In

2003, the sovereign bond interest spread of South African government bonds to USA bonds was 1.75%, in 2005 it was 1%, in 2010 it was 1.4% and in 2014, the spread had increased to 2.4%. S&P(in 2017) and Moody's (in 2020) downgraded South Africa's sovereign credit rating to below investment grade and by October 2023, the interest spread between South Africa's government debt (Rands) and the USA had widened to 7.78%.

Leases

Leases are a common source of finance for the funding of movable assets. Leases, like loans, can be structured in almost any way. They are secured as the lessor retains ownership of the Page 13-30 asset, while the lessee has the use of the asset during its useful life. The lease payments are computed so as to offer the lender the cost of the asset plus a reasonable return on the funds. Leases are discussed fully in Chapter 15.

Short-term debt

Short-term finance is usually raised in line with the working capital cycle of the business. Many businesses are affected by seasonality, which means that sales do not occur evenly over time. Rather, there are peaks and troughs of activity. In such cases short-term debt is frequently used as a form of bridging finance to bolster the company's financial resources as it builds up its production and/or inventories prior to sale, and debtors prior to receiving settlement.

Much short-term finance is raised spontaneously through trade creditors. It is nevertheless a matter of policy as to whether the company wants to use short-term or some other form of finance. If it chooses to do so, it is exposed to higher risk as short-term finance needs to be met sooner than long-term finance, or it needs to be renegotiated more frequently.

Short-term finance is discussed in the chapter dealing with working capital, but some of the more common sources of short-term finance are mentioned here:

- *Bank overdraft.* Bank overdraft is the facility to make payments beyond the amount of money in the bank account. The intention of the bank overdraft is to provide a form of bridging finance to tide the company through its working capital cycle. In South Africa it is common practice that part of the overdraft is, in reality, a source of medium-term finance.

- *Banker's acceptance*. A banker's acceptance (BA) is created when the company sells a bill of exchange to the bank to be settled on a predetermined date, usually 90 days later. It is cheaper than a bank overdraft, but it does not have the same degree of flexibility. The company is committed to using the full amount of banker's acceptance for the agreed period, whereas the overdraft can be managed more easily. BAs are used mainly to fund seasonal fluctuations.
- *Bills and notes*. There are bills and notes other than the banker's acceptance; however, owing to the ease of dealing in bankers' acceptances and their quality (drawn on a bank) these other instruments are less frequently used.
- *Debtor Finance.* Banks will provide short-term financing against the security of a company's debtors - normally up to 75% of the value of a firm's debtors' invoices.
- *Revolving Credit Facility.* A company is required to make fixed monthly instalments but can withdraw funds up to the original limit once 20%-30% of the amount has been repaid.

See Chapter 12 for more on the use of short-term debt to finance working capital.

5 HYBRID INSTRUMENTS

Hybrids are combinations of debt and equity. Simple forms of these were mentioned in relation to preference shares and debentures. Hybrids may also be created by combining elements of the spot and derivative markets. The reason for their existence is that they can be specially constructed to suit the needs of the situation.

Classifying hybrid instruments as debt or equity is difficult. Pure debt and pure equity instruments are at opposite ends of a continuum. Between these extremes there are many instruments which are a combination of debt and equity. The following may help in classifying these as debt or equity:

- Equity has a residual right to the wealth of the firm, whereas debt has a contractual right. This means that the equity holders have a right to what is left over after all other stakeholders, including debt holders, have been satisfied. Thus instruments that have a contractual right to income, capital, or priority in times of difficulty would tend to be debt.
- Payments for the use of debt are usually tax deductible, whereas those for equity are not.

- Debt tends to have a finite life. The life of equity tends to be the life of the firm.
- Equity holders control the firm, debt holders do not.

6 COMPARISON OF DEBT AND EQUITY

The relationship between debt and equity is referred to as the capital structure. Capital structure is discussed more fully in Chapter 14. In making the decision between raising finance through debt or equity, one of the basic principles of finance comes into consideration: there is always a trade-off between risk and return. Therefore, as risk is increased through the use of higher proportions of debt, so an increase in return is expected. The use of debt increases the risk as the interest payments have to be met, regardless of the financial well-being of the company.

In the following section, certain features of debt and equity will be highlighted and comment will be passed as to when they may be advantageous or disadvantageous. These will be discussed under the headings of return, risk, and control. For a fuller discussion on risk and return in relation to the use of debt and equity refer to Chapter 14.

Return

Specified interest

The company must pay the interest agreed upon with the lender, irrespective of the performance of the company. This is an advantage when the company is doing well and earning more than the interest charge. In such circumstances the effect is to lever the profits so that the return on shareholders' funds is higher than the return on assets. On the other hand, the reverse is true when the return on assets is lower than the interest charge. In these circumstances the return to the shareholders is lower than the return on assets. This is known as negative gearing. If the situation persists, the company could go insolvent.

Tax-deductibility

The interest charge on debt is usually tax-deductible. Its effective cost to the company is therefore the after-tax cost of debt. Dividends, on the other hand, are not tax-deductible and so there is no tax relief from the company's point of view.

Cost

The cost of debt is usually less than the cost of equity as a result of the lower risk to the lender and the tax deductibility of the interest. The use of debt allows the company to lever the return to shareholders as described under 'specified interest' above. Furthermore, the costs involved in issuing ordinary shares are higher than the costs of issuing debt.

Risk

Commitment

Interest payments on debt must be met whether there are profits or not. In times of economic hardship this can be an onerous requirement. A company is not required to pay a dividend to ordinary or preference shareholders.

Capital repayment

Debt requires a capital repayment. Ultimately all debt must be repaid, so the company must not only earn sufficient profits for the interest payments, but it must also make adequate provision for the capital repayments. This may put a strain on the liquidity of the company. Ordinary share capital is not required to be repaid to the shareholders as it forms part of the firm's permanent capital. This means that there will be no burden placed on the liquidity of the company arising from the issue of ordinary shares. Page 13-32

Capital structure

There is a limit to the amount of debt that can be raised before the market reassesses the company's risk profile, which will be reflected in the cost of capital. The equity holders will require a greater return on their investment and, similarly, it will be possible to raise more debt only from marginal lenders who will charge higher interest rates and/or impose other restrictions on the company. You are referred to the discussion on breaks in the marginal cost of capital curve in Chapter 7.

The more debt the company raises, the closer it comes to this saturation point. Consequently, raising debt reduces the company's flexibility for raising future finance. This can be to the company's disadvantage if debt becomes available on favourable terms.

Signalling effects

The sale of new ordinary shares may be perceived to be a negative signal by investors as it may indicate management's view that the ordinary shares are overvalued at current levels and therefore represents an optimal time to undertake an issue of ordinary shares. On average, rights issues have resulted in a fall in the share price of 2–5%. The issue of debt may indicate the opposite view.

Control

Dilution of control

Raising finance through the issue of equity dilutes the control of the existing shareholders. If the company needs to raise equity finance over and above the amount of the earnings that can be retained, then it will need to issue new shares. In such circumstances, if the existing shareholders do not want to take up new shares, or cannot do so, then their holding in the company will be diluted, and so they will have given up a measure of the control they had. The ability to control a company is of value, so they would not want to relinquish control lightly.

Raising finance through debt can also have control implications. For example, the provider of debt may impose certain restrictions on the company as a condition of the loan. These restrictive covenants are usually designed to limit the opportunity for management to take actions, which could place the debt holder at a disadvantage. These could include restrictions on the ability to pay dividends, dispose of assets or maintaining certain working capital ratios. The debt provider may also demand a seat on the board or a regular report on the performance of the company.

Levels of control

Control over the affairs of a company is the ability to make operating, investment and financing decisions. There are, however, many different levels or degrees of control, and the crossing from each of these levels to another affects shareholder value.
Some of the more important points of control are:

- *75% of issued share capital*. This enables the passing of special resolutions.
- *50% + 1 share of issued share capital*. This enables the passing of ordinary resolutions.
- *35% of issued share capital*. If an investor acquires 35% of the ordinary shares, then a similar offer must be made to all shareholders in terms of the Companies

Act.

- *20% to 35% of issued share capital*. It is generally believed that this is enough to control a listed company depending on how the other shares are held.
- *Control of the board*. Control may be exercised not only through the holding of shares, but also by the ability to control or to have influence over the board of directors.
- *Default.* Control may be lost through the company's failing to meet certain commitments which transfer certain rights to other parties. For example, if a company falls in arrears on the payment of its preference dividend, the preference shareholders are granted voting rights. This may cause the relative holding of the ordinary shareholders to fall below one of the control levels mentioned. Page 13-33

FROM THE REAL WORLD – CORPORATE BONDS AND BORROWINGS

SASOL

SASOL operates mainly in Southern Africa but also operates in the USA. In May 2023, SASOL reported that it had issued a US$1bn (R18.8bn) bond with a fixed interest rate of 8.75%, due in 2029. Included in SASOL's borrowings at 30 June 2023 are the US$3.8bn (R70.6bn) bonds with fixed interest rates of between 4.38% and 6.5% which are listed on the NYSE. In terms of local borrowings in October 2022, SASOL issued domestic medium term notes (DMTN) of R2.066bn as part of its R15bn DMTN programme. SASOL has also issued senior unsecured convertible bonds and also obtained a revolving credit facility of US$1.987bn and a term loan of US$982m. The US bond yield in 2023 of 8.75% was much higher than in 2012 when SASOL had managed to issue bonds in the USA at a coupon rate of 4.5% (the lowest coupon achieved by a South African company) and this increase in rates reflects the increase in interest rates and credit risk spreads facing South African firms. In 2023, SASOL reported an objective to reduce debt financing to less than 1.5× Net Debt: EBITDA and get Net Debt below US$4bn. The weaker closing Rand/US$ exchange rate resulted in a R16bn increase in debt in FY2023.

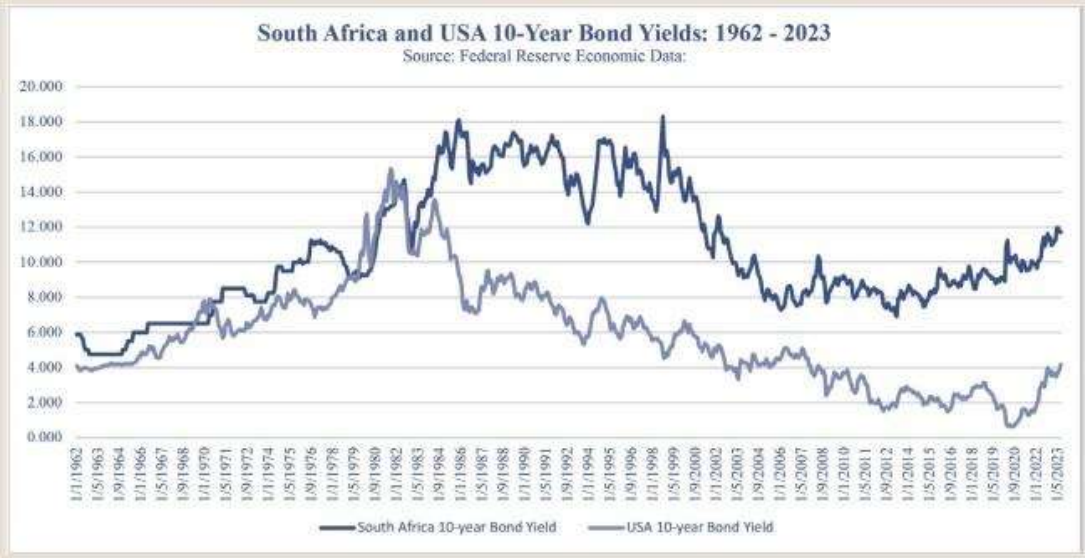
SASOL in 2023 also reported the following:

> Our debt is comprised of different instrument notes, which by their nature either bear interest at a floating or a fixed rate. We monitor the ratio of floating and fixed interest in our loan portfolio and manage this ratio, by electing to incur either bank loans, bearing a floating interest rate, or bonds, which bear a fixed interest rate. We may also use interest rate swaps, where appropriate, to convert some of our debt into either floating or fixed rate debt to manage the composition of our portfolio.

The CFO stated that SASOL executed debt refinancing transactions to address the 2024 maturities and thereby extend the maturities of the debt. In FY2023, 77% of debt was fixed rate debt and 23% was variable rate debt.

SOUTH AFRICAN & USA 10-YEAR BOND YIELDS

The following graph sets out the 10-year bond yields for South African Rand currency government bonds in relation to the 10-year bond yields for USA government bonds. It clearly depicts the higher cost of borrowing in South Africa, which reflects higher inflation and differences in sovereign credit risk. This raises the cost of investment and financing decisions for South African companies and government entities. Page 13-34



PPC

Expansion into the DRC combined with difficult operating and economic conditions as well as high debt levels resulted in the PPC share price falling from around R30 per share in 2014 to R6 in 2016. This required that PPC make onerous financing decisions to ensure that it would be able to survive. Firstly, it undertook a R4bn rights issue to bolster its equity and reduce debt, whereby it issued 1bn shares at R4 per share but incurred transaction costs of R189m. At the beginning of the year, there were about 607m ordinary shares in issue. The rights issue (and the unwinding of BEE I) ensured that the net debt/EBITDA ratio declined from 3.7× to 2.3× which reflected a fall in net debt from R8.7bn in 2016 to R4.7bn in 2017. The company also improved its debt maturity profile (which reflects when it has to repay its loans) from 2017 to 2018 and later to 2023 – but at a significant cost. PPC stated the following in its 2018 annual financial statements:

> In order to further strengthen the group's liquidity position and maturity profile, the South African debt was successfully renegotiated in March 2018. A R2 billion facility was secured, with an extended maturity profile at reduced interest rates. In the DRC, management was successful in negotiating a two- year capital repayment grace period with interest still required to be settled during this two-year period. The interest rate has increased to six-month LIBOR plus 9.75%.

This is a high cost of borrowing (in US Dollar terms) and reflects the impact of refinancing debt when under financial distress.

By 2023, PPC had restructured its investment and loans in the DRC with the IFC and other lenders thereby removing a significant debt burden. DRC is now only an equity investment. Total borrowings have fallen from R4682m in 2018 to R1189m and this is reflected in the graph above.

7 INFLATION-LINKED BONDS

Page 13-35 Governments, state entities, banks and large corporate firms are increasingly considering the issue of inflation-linked bonds ("linkers"), which will offer investors a hedge against inflation. Investors protect the purchasing power of their investments over time. How do inflation-linked bonds work? The bond pays a fixed real coupon rate. The nominal amount (principal) and the coupon payment increase with inflation. A brief example will indicate how an inflation-linked bond works. Assume that the RSA government issues an inflation-linked bond with a real coupon rate of 2%, which will be redeemed in 3 years' time. Inflation is expected to be 6% per year for the next 3 years. Assume the initial principal amount is R1m and interest is payable in arrears at the end of each year. The cash flows and values will be as follows:

Year	Inflation rate	Cumulative inflation factor	Nominal value of Principal	Nominal Interest Payment	Real Interest Payment
20x5	6%	106.0000%	1,060,000	21,200	20,000
20x6	6%	112.3600%	1,123,600	22,472	20,000
20x7	6%	119.1016%	1,191,016	23,820	20,000

Note: Cumulative inflation factor: 20x6: (1.06)(1.06) = 1.1236, 20x7: (1.06)(1.06)(1.06) = 1.191016

The real value of the nominal interest payment will amount to R20 000 each year. The nominal redemption value of R1 191 016 at the end of 20x7 will have a real value of R1m [1191016/1.191]. If the inflation rate changes in 20x7, to say 5%, then the nominal interest payment in 20x7 would reduce to R23 596 and the nominal redemption value would fall to R1 179 780.[6]

Assume that it is estimated that the future inflation rate is 6%. The company could have issued straight bonds with a quoted coupon rate of 8.12%. Assume that the actual inflation rate turns out to be 6% each year. Then, if we issued straight bonds or inflation-linked bonds, the YTM for both bonds would be the same.

Inflation-linked	20x4	20x5	20x6	20x7
Principal	(1,000,000)	21,200	22,472	1,191,016
Coupon pmt				23,820
	(1,000,000)	21,200	22,472	1,214,836
YTM (IRR)	8.120%			
Straight	**20x4**	**20x5**	**20x6**	**20x7**
Principal	(1,000,000)	81,200	81,200	1,000,000
Coupon pmt				81,200
	(1,000,000)	81,200	81,200	1,081,200
YTM (IRR)	8.120%			

However, we will now assume that right after the issue of the bonds, the inflation rate increases to 10% per year. The straight bonds will continue to have a yield of 8.12%, which is lower than the inflation rate (and which will result in a fall in the value of the bonds) while the YTM of the inflation-linked bonds will increase to 12.2%.

Inflation-linked	20x4	20x5	20x6	20x7
Principal	(1,000,000)			1,331,000
Coupon pmt		22,000	24,200	26,620
	(1,000,000)	22,000	24,200	1,357,620
YTM (IRR)	12.200%			

For investors, a government bond while risk-free in relation to the repayment of capital, carries significant inflation risk while a government inflation-linked bond protects investors from the risk that higher inflation will reduce the real value of bonds and the real value of the bond's cash flows. An inflation-linked government bond reflects more closely what we call a riskless asset. Some companies are able to increase prices with inflation and are therefore effectively hedged if they issue inflation-linked bonds. Pension funds will often have inflation-linked liabilities and may wish to protect the real value of their investments by investing in inflation-linked